Architectural and Operational Blueprint for the Bandh Prahari 3D Flood Management Platform in Bihar

The management of flood regimes in Northern Bihar represents one of the most structurally complex hydrological, engineering, and administrative challenges globally. The region's flood dynamics are dictated by massive Himalayan river systems—including the Kosi, Gandak, Bagmati-Adhwara, Kamla-Balan, and Mahananda—which transport extreme volumes of water and heavy silt loads from their upstream catchments in Nepal directly into the flat Gangetic plains. During periods of intense precipitation in the Himalayan catchments, barrages such as Valmikinagar on the Gandak (design capacity of ~8.5 lakh cusecs) and Birpur on the Kosi (design capacity of ~9.5 lakh cusecs) are frequently forced to release sustained high discharges. These releases, which routinely exceed 4 to 5.5 lakh cusecs for durations spanning 10 to 30 hours, cause downstream river gauges to rapidly surpass both Danger Levels (DL) and Highest Flood Levels (HFL).

To protect the ~68,800 square kilometers of flood-prone area—constituting 73% of Bihar's total geographical area and placing 76% of the North Bihar population at recurring risk—the state relies on a vast defensive network of approximately 3,750 kilometers of earthen embankments, known locally as Tatbandhs. Despite continuous anti-erosion works and asset monitoring, these structures fail almost annually. Documented failures during the 2026 monsoon events, such as the Gandak breaches at Sheetalpur in Gopalganj, Haijalpur in Saran, and various sites in West Champaran, illustrate a persistent failure cascade: sudden upstream rainfall leads to barrage surges, generating extreme hydrostatic pressure that induces localized geotechnical failures, ultimately resulting in the rapid inundation of the surrounding countryside.

The technical blueprint for the "Bandh Prahari" platform proposes a production-grade, 3D flood monitoring, prediction, and relief management system hosted on Amazon Web Services (AWS). Conceived as a highly integrated codebase spanning a modern MERN stack frontend, a Python-driven machine learning backend, and serverless orchestration, the platform is designed not to replace but to tightly augment the existing digital infrastructure of the Bihar Government. This includes bi-directional integration with the Flood Management Improvement Support Centre (FMISC), the Bihar Embankment Asset Management System (BEAMS), the Real-Time Data Acquisition System (RTDAS), the State Emergency Operation Centre (SEOC), and the Aapda Sampoorti Direct Benefit Transfer (DBT) portal managed by the National Informatics Centre (NIC).

This report provides an exhaustive analysis of the Bandh Prahari architectural blueprint, evaluating its integration with local Bihar government infrastructure, its capacity to bridge critical operational gaps at the granular field level, its empowerment of local volunteer forces using real-time 3D data, and the intricate technical mechanisms required to connect global cloud infrastructure to local administrative bodies.

Deconstructing Systemic Gaps in Bihar’s Flood Management Ecosystem

To engineer an effective augmentation platform, it is necessary to thoroughly analyze the existing gaps within Bihar's flood management framework. These systemic vulnerabilities span socio-economic assessment methodologies, geotechnical surveillance latencies, and topographical resolution constraints.

The Assessment-Based Vulnerability Paradox

Current disaster assessment and compensation frameworks in Bihar frequently fail to capture the true multidimensional impact of floods on marginalized communities. A comprehensive 2024 household-level flood loss assessment conducted by the Megh Pyne Abhiyan (MPA) and Tata Trusts evaluated the impact across 2,290 households in 134 wards spanning seven severely affected districts in North Bihar. The assessment recorded total economic losses of ₹126.3 crore in a single flood phase, with land damage accounting for 47.2% of the losses and housing damage accounting for 36.2%.

The data revealed a critical phenomenon identified as the "assessment-based vulnerability paradox". Standard official compensation frameworks are heavily tied to the absolute monetary value of damaged assets. Consequently, asset-rich households (predominantly from the General category) reported the highest absolute monetary losses, with average household damages approaching ₹9.8 lakh. In contrast, asset-poor households, frequently belonging to Scheduled Caste (SC) or Scheduled Tribe (ST) demographics, reported significantly lower absolute monetary losses. However, standard metrics obscure the reality that for asset-poor households, the loss of a minimal number of assets represents a catastrophic shock that strips away their entire livelihood base, triggering severe distress borrowing and food insecurity.

The existing relief mechanisms, reliant on post-disaster manual surveys, lack the real-time spatial granularity to target the most structurally vulnerable populations swiftly, resulting in prolonged delays in the disbursement of direct benefit transfers. The Bandh Prahari platform addresses this by automating the verification and disbursement pipeline through offline-first digital ledgers directly linked to government portals.

Demographic / Asset Profile	Average Absolute Loss Reported	Impact on Livelihood Base	Coping Mechanisms Observed
Asset-Rich (General Category)	₹1.2 lakh to ₹9.3 lakh	High monetary loss, moderate structural resilience	Utilization of savings, formal credit access
Asset-Poor (SC/ST/Marginalized)	Below ₹1.0 lakh	Catastrophic loss of primary livelihood tools	Distress asset sales, high-interest informal borrowing, forced migration

The Latency in Embankment Maintenance and Surveillance

The Water Resources Department (WRD) of Bihar operates the BEAMS portal, which serves as a centralized inventory of embankment assets, chainages, and structural conditions across the state's river basins. The physical surveillance of these ~3,750 km of embankments during the peak monsoon season relies entirely on manual patrols executed by the WRD's Flood Fighting Force, which is subdivided into 14 distinct zonal divisions.

Geotechnically, earthen embankments exhibit highly visible precursor signals hours or even days prior to a catastrophic structural collapse. The primary mechanism of failure is internal erosion, commonly referred to as piping, which manifests on the landward side of the

embankment as sand boils. When a sand boil discharges highly turbid or muddy water, it indicates that the internal foundation material of the earthen core is being actively eroded and transported out by the high hydraulic gradient, signaling an imminent failure.
The critical operational gap in Bihar is not a lack of observation, but the extreme latency between the physical manifestation of these signals and the deployment of an accountable engineering response. Local citizens, patrol workers, or farmers frequently observe these symptoms, but the transmission of this intelligence through the official administrative hierarchy is slow. This latency routinely results in false acknowledgments or delayed interventions, allowing a manageable localized piping issue to evolve into a massive uncontainable breach.

The Topographical Resolution Deficit in Predictive Modeling

Accurate predictive flood modeling requires high-fidelity topographical data. However, the standard Digital Elevation Models (DEMs) widely available for the region, such as the Copernicus 30m or the SRTM 30m, lack the spatial resolution required to resolve Bihar's river embankments accurately. Embankment crests in the region typically measure only 4 to 8 meters in width. When processed through a 30m raster grid, the elevation of the narrow embankment crest is mathematically averaged with the significantly lower elevation of the adjacent floodplain. This artificial "flattening" of the protective barriers severely corrupts 2D hydraulic models. Simulated floodwaters mathematically overtop the flattened embankments at much lower discharge thresholds than in reality, generating highly inaccurate inundation extents and false alarms. Furthermore, the Kosi river basin is characterized by extreme sediment dynamics; of the roughly 100 million tonnes of sediment entering at Chatara annually, massive quantities are deposited within the embanked channel, resulting in the progressive superelevation of the riverbed above the surrounding topography. Standard DEMs fail to dynamically capture this aggradation, rendering standard flood models obsolete in the face of rapidly altering channel morphology.

Bridging Global AWS Infrastructure with Local Bihar Systems

The architectural philosophy of the Bandh Prahari platform is centered on non-replacement. It acts as an intelligent, high-speed middleware layer that tightly couples massive global cloud computing power with the existing, localized digital infrastructure of the Bihar government. Based on the developer profile associated with the codebase, the system leverages a modern full-stack implementation: a robust Node.js/Express application programming interface (API) layer handling concurrent spatial requests, a React-based frontend for administrative dashboards, and Python-driven machine learning microservices deployed on AWS for heavy geoprocessing tasks.
To meet the rigorous compliance and security standards mandated by the Indian government, the entire AWS infrastructure is deployed strictly within Ministry of Electronics and Information Technology (MeitY) empanelled regions. The primary architecture resides in the AWS Asia Pacific (Mumbai) region (ap-south-1), with a disaster recovery environment maintained in the Hyderabad region (ap-south-2).
The connection between the AWS cloud environment and the local Bihar infrastructure relies on highly secured transit pathways. Communication with state-owned endpoints, such as the FMISC MapServer, BEAMS APIs, and the Aapda Sampoorti portal, is secured using Mutual

Transport Layer Security (mTLS) on Amazon API Gateway. This ensures that only authorized, certificate-bearing client services can ingest sensitive telemetry or push enriched incident data back into the state's official dashboards. Furthermore, in strict adherence to the directives issued by the Indian Computer Emergency Response Team (CERT-In), the platform implements a Write-Once-Read-Many (WORM) storage architecture. All system incident logs, escalation events, and API access records are encrypted using AWS Key Management Service (KMS) and retained in an Amazon S3 bucket configured with Object Lock for a mandatory 180-day rolling period, ensuring immutable forensic auditability.

Bihar Government Entity / System	AWS Integration Component	Primary Function	Integration Protocol / Security
FMISC & BEAMS	Amazon API Gateway / AWS Lambda	Bi-directional sync of gauge levels, asset chainage, and enriched breach incident data.	REST APIs secured via mTLS
State Emergency Operation Centre (SEOC)	EventBridge / API Gateway	Automated generation and routing of CAP 1.2 XML alerts for cell-broadcast siren triggering.	Secure Webhooks / XML
Disaster Management Department (DMD)	Amazon Bedrock / S3	Automated compilation of IRS Form 001 Incident Briefing PDFs for District Magistrates.	Signed URLs / Bedrock Agents
Aapda Sampoorti (NIC Bihar)	AWS AppSync / DynamoDB	Export of verified, deduplicated beneficiary lists for ₹7,000 DBT PFMS payouts.	JSON/CSV Export matching NIC schema

Deep Data Ingestion and Normalization Architecture

Real-world flood prediction engines fail when they assume that raw data from the field arrives cleanly. The Bandh Prahari platform relies on a highly resilient ingestion pipeline capable of harmonizing unstructured crowdsourced media with structured official telemetry.

The platform ingests official state and central telemetry through automated AWS Lambda scrapers and API clients. These scrapers operate at aggressive 15-to-30-minute intervals during the active monsoon season, extracting dynamic hydrology data from the FMISC MapServer, WRD barrage release tables, and Central Water Commission (CWC) forecasting stations. Additionally, the pipeline continuously monitors the NDMA SACHET RSS feeds, decoding Common Alerting Protocol (CAP) 1.2 XML payloads related to sudden upstream barrage discharges. All structured data streams are routed into Amazon Timestream, a purpose-built time-series database configured with a 14-day memory store to facilitate ultra-low latency queries for the predictive engines, and a 10-year magnetic store to retain historical flood trends for post-monsoon analysis. A unified schema normalizes disparately formatted FMISC station codes, CWC identifiers, and Local Government Directory (LGD) village codes, ensuring that downstream systems never mix critical spatial identifiers.

Concurrently, the platform manages unstructured crowdsourced reports from local citizens and WRD patrol workers. Users interact with the system via a localized WhatsApp bot or a

lightweight Android client, submitting GPS check-ins alongside voice notes recorded in regional dialects (Hindi, Bhojpuri, Maithili) and photographs of the embankments. To prevent the API Gateway from becoming a bottleneck during viral traffic surges associated with a breach event, binary media uploads bypass the gateway entirely. Instead, the mobile client requests a single-use Amazon S3 presigned URL, allowing the device to upload massive photo and audio payloads directly to a raw ingestion bucket.

The extraction of actionable intelligence from these regional audio files relies on an advanced generative AI pipeline. Upon the upload of a voice note, an S3 event trigger invokes Amazon Transcribe, specifically configured for the hi-IN localization, to generate a raw text transcript. This transcript is subsequently routed to Amazon Bedrock, utilizing advanced Large Language Models (LLMs) such as Anthropic Claude or Amazon Nova. By leveraging the Bedrock Converse API's tool-calling schema, the LLM is restricted from generating conversational output; instead, it is forced to extract specific variables into a strict JSON format. The model evaluates the transcript to identify the specific embankment chainage (e.g., "km 46.5"), classify the physical symptom (e.g., BOIL, PIPING, SLUMP), and ascertain the water clarity (TURBID_MUDDY versus CLEAR). To optimize performance and reduce inference costs, the implementation utilizes prompt caching strategies, specifically caching the large tool definitions using the TOOLS_ONLY or SYSTEM_AND_TOOLS cache breakpoint configuration.

Finally, the ingestion layer incorporates an automated satellite remote sensing pipeline. An Amazon EventBridge rule monitors the AWS Open Data Registry for new acquisitions of Sentinel-1 Ground Range Detected (GRD) synthetic aperture radar (SAR) imagery over the relative orbits covering the Kosi and Gandak basins. SAR imagery is vital because optical satellites are blinded by dense cloud cover during the monsoon. The notification triggers an AWS Batch containerized job that executes radiometric calibration, Lee speckle filtering, and dynamic Otsu thresholding on the VV/VH polarized backscatter coefficients to delineate open water surfaces from dry land. The resulting Cloud Optimized GeoTIFFs (COGs) are pushed to S3, complementing the FMISC's existing RADARSAT/NRSC inundation products.

Predictive AI and USACE-Aligned Breach Scoring

The intelligence core of the Bandh Prahari platform utilizes three specialized, decoupled analytical modules rather than a single, opaque machine learning model.

The first module focuses on discharge and stage forecasting. It employs a Spatio-Temporal Graph Neural Network (ST-GNN) combined with Bidirectional LSTM units deployed on Amazon SageMaker Serverless Inference. The architecture models the river network as a graph, where nodes represent major gauges along the Kosi, Gandak, and Bagmati basins, and edges represent the flow connectivity and reach lengths. The model ingests multi-dimensional feature sets, including high-resolution IMD quantitative precipitation forecasts (GFS/WRF grids), rainfall data from the upstream Nepal catchments, scheduled barrage release volumes, and the 7-day rolling discharge histories pulled from Amazon Timestream. The output is a highly accurate 24-to-72-hour forecast of water stage levels and discharge rates at key downstream stations. The serverless deployment strategy ensures that the compute cluster scales down to zero during the dry winter and summer months, eliminating idle infrastructure costs.

The second module is the heart of the tactical response: a deterministic embankment vulnerability scoring engine. Rather than relying purely on predictive AI, this engine codifies the rigorous geotechnical standards established by the US Army Corps of Engineers (USACE) Flood Fight Handbook, calibrated specifically with local weights derived from documented Bihar

failure precursors. The engine calculates a continuous Risk Score based on the formula: \text{Risk Score} = w_1 \cdot S_{\text{sign}} + w_2 \cdot \Delta H_{\text{gauge}} + w_3 \cdot V_{\text{history}} + w_4 \cdot T_{\text{saturation}}
Here, S_{\text{sign}} evaluates the severity of the observed physical symptom, \Delta H_{\text{gauge}} represents the current hydrostatic head relative to the danger level, V_{\text{history}} factors in the historical vulnerability of that specific chainage pulled from the BEAMS database, and T_{\text{saturation}} measures the duration of continuous saturation.
The strict categorization rules are rooted in the physical mechanics of internal erosion. According to geotechnical principles, a "quick" or boiling condition occurs when the upward seepage force of water passing through the embankment foundation equals the buoyant weight of the soil particles. This critical hydraulic gradient (i_{cr}) is defined by Terzaghi's equation: i_{cr} = \frac{G_s - 1}{1 + e}
Where G_s is the specific gravity of the soil solids and e is the void ratio. When a local worker reports a sand boil discharging turbid or muddy water, the USACE engine recognizes that the critical hydraulic gradient has been exceeded, and the fine core material of the embankment is actively being transported out, drastically reducing the overall factor of safety. The engine deterministically classifies this as a Priority 1 (Extreme Immediate Hazard) incident, bypassing lower-level evaluations. Conversely, a sand boil discharging clear water indicates a high pressure state without active material loss, which is classified as a Priority 2 (Severe Risk) requiring continuous monitoring. Any physical slump or slope slip occurring while the river gauge is at or above the Danger Level is immediately escalated to P1, as the loss of resisting soil mass against extreme hydrostatic pressure guarantees imminent failure.

Geotechnical Symptom	Water Clarity / Condition	USACE Engine Priority	Structural Implication
Sand Boil / Piping	Turbid or Muddy	**P1**	Active internal erosion; core foundation material is being expelled, precipitating collapse.
Sand Boil / Piping	Clear	**P2**	High hydraulic gradient present; soil matrix remains intact but requires immediate counter-pressure (sandbag ringing).
Overtopping	Gauge > DL + 0.5m	**P1**	Immediate erosion of the downstream slope crest leading to rapid breach widening.
Slope Slump / Slip	Gauge \ge DL	**P1**	Critical loss of stabilizing mass on the landward slope.
Seepage	Saturation > 48 hours	**P2**	Prolonged saturation diminishes effective stress and the overall shear strength of the earthen structure.

The third module automates geospatial inundation segmentation. When the USACE engine logs a P1 incident or an actual breach is confirmed, a SageMaker Geospatial container is triggered. It computes a hydrodynamic 2D shallow-water approximation using a D8 flow-routing algorithm processed over a highly customized hybrid DEM (where high-resolution embankment breaklines are digitally burned into the terrain model). The resulting inundation boundary polygon is immediately routed to the Amazon RDS PostGIS instance. The database executes an ST_Intersects spatial query against the Census 2011/2021 Local Government Directory (LGD) layer, instantly returning a deterministic array of affected villages, vulnerable ward populations, and specific panchayats that lie within the projected path of the floodwaters.

The "Jawab Ghadi" Accountability Engine: Enforcing Official Response

The most sophisticated predictive models are rendered useless if the intelligence is not translated into immediate, accountable physical intervention. Historically, the transmission of threat data through the Bihar WRD hierarchy has suffered from severe latency, exacerbated by a lack of verifiable accountability mechanisms at the field level. The Bandh Prahari blueprint eliminates this operational gap through the "Jawab Ghadi" (Answer Clock) escalation engine, a deterministic workflow powered by AWS Step Functions.

Traditional application-level timers managed in system memory are brittle; they are prone to failure during server crashes or redeployments, making them unsuitable for critical disaster management operations. AWS Step Functions Standard Workflows mitigate this by providing persistent, crash-proof execution states that can wait for extended durations.

When the USACE engine categorizes an incident as P1, the Step Functions state machine executes an AWS Lambda task that dispatches an actionable, interactive WhatsApp template directly to the specific Junior Engineer (JE) assigned to that WRD Flood Fighting Force zone. Crucially, the Lambda function invokes this task utilizing the .waitForTaskToken integration pattern. The Step Functions service generates a unique TaskToken, which is stored alongside the incident metadata in Amazon DynamoDB. The state machine then enters a strict Wait state configured with a TimeoutSeconds value of 600 (10 minutes).

If the JE interacts with the WhatsApp message by tapping the "On My Way" button, the webhook payload is received by the API Gateway. A subsequent Lambda function retrieves the stored TaskToken and issues a SendTaskSuccess API call back to Step Functions, allowing the state machine to transition smoothly to the success branch. However, if the 600-second timer expires without the service receiving the callback, the Catch block is triggered immediately. The system generates an immutable non-compliance audit record stored in the S3 Object Lock bucket. It then automatically escalates the incident by paging the Assistant Engineer (AE) and simultaneously copying the Executive Engineer (EE) and the District Magistrate (DM), effectively forcing hierarchical intervention and bypassing local communication blockages.

To combat the pervasive issue of false acknowledgments—where personnel remotely acknowledge an alert without physically attending the site—the Jawab Ghadi engine integrates Amazon Location Service for presence verification. Upon a successful initial response from the JE, the workflow transitions to a 45-minute geofence enforcement state. The system dynamically generates a 200-meter circular geofence around the precise spatial coordinates of the reported embankment incident. The responding engineer's mobile application must securely stream GPS location check-ins via MQTT over AWS IoT Core. The Location Service evaluates these coordinates; only an ENTER event registered within the dynamic geofence satisfies the

presence requirement. If the 45-minute window elapses without an ENTER event, the system escalates the breach site status to "Unmanned Emergency" and triggers high-priority alarms at the SEOC in Patna.
Furthermore, the Jawab Ghadi engine manages automated night roll-calls, addressing the heightened vulnerability of embankments during nighttime hours when manual surveillance naturally wanes. Utilizing Amazon EventBridge Scheduler, the system initiates recurring checks whenever an upstream CWC gauge crosses the Danger Level. Between the hours of 22:00 and 06:00, interactive WhatsApp prompts are pushed to designated patrol workers every two hours, requiring them to confirm the status of their specific kilometer stretch ("km X: Sab theek?"). Failure to respond within 20 minutes automatically flags that specific chainage on the FMISC dashboard as an unmonitored zone, instantly notifying the JE to dispatch a motorized inspection unit.

3D Geospatial Visualization for Local Governance and Volunteers

Standard 2D WebGIS mapping portals are wholly insufficient for tactical flood response in areas characterized by massive sediment aggradation. Decision-makers and local volunteer forces cannot intuitively grasp the volumetric threat posed by a river flowing meters above the elevation of the adjacent villages solely by looking at flat polygons. The Bandh Prahari platform mandates a highly optimized 3D visualization layer, engineered specifically to render complex terrain and fluid dynamics smoothly on low-cost Android smartphones operating over congested 2G and 3G networks prevalent in rural districts.
To overcome the inherent resolution deficits of standard satellite DEMs, the platform employs a server-side mesh fusion architecture. Baseline terrain data is ingested alongside the high-resolution embankment vector shapefiles (detailing chainage lines, crest heights, and toe lines) maintained in the WRD BEAMS database. An AWS Batch worker utilizing the Cesium Terrain Builder application triangulates the raw DEM into a Triangulated Irregular Network (TIN). The algorithm then "burns" the embankment vector breaklines directly into the TIN, forcing the triangular vertices of the mesh to align perfectly with the surveyed physical crest elevation. The resulting data is exported in the highly efficient OGC Quantized Mesh format (.terrain), structured in standard Slippy Map directories, and hosted on an Amazon S3 origin. Concurrently, building footprints, raised evacuation platforms, and schools are extracted from OpenStreetMap (OSM) and processed via py3dtiles to generate Batched 3D Models (.b3dm), which are semantically enriched with official Local Government Directory (LGD) metadata.
Rendering massive, dynamic flood surfaces using traditional pre-computed raster tiles consumes vast amounts of storage and network bandwidth, rendering the application unusable in rural environments. Instead, the platform utilizes client-side WebGL and GLSL shaders (deployed via CesiumJS or MapLibre GL Native). The application fetches the predicted Water Surface Elevation (WSE) from the Amazon Timestream API. The client-side shader instantiates a flat, dynamic water surface plane positioned at the precise H_{WSE}. The GLSL fragment shader executes a per-pixel depth comparison against the loaded Quantized Mesh geometry; pixels where the terrain elevation is mathematically lower than the water elevation are rendered using a translucent blue water shader featuring specular reflection and wave-normal distortion. Where the terrain exceeds the water height, the raw satellite texture remains visible. This approach decouples the hydraulic model from the 3D rendering pipeline, achieving massive reductions in payload size.

To guarantee fluent loading over constrained networks, all 3D assets are distributed via Amazon CloudFront configured with Origin Access Control (OAC). Static terrain (.terrain) and village assets (.b3dm) are served with immutable, long-term cache headers (max-age=31536000), allowing mobile devices to cache the gigabytes of mesh data locally. The live API endpoints dictating the water elevation and breach status are served with extremely short time-to-live settings (max-age=15). The implementation of Brotli compression on the CloudFront edge nodes further reduces the 3D tile payloads by up to 35%.

Relief Logistics and Aapda Sampoorti DBT Integration

When an embankment breaches and inundation commences, the platform's operational focus pivots instantly from geotechnical monitoring to humanitarian relief logistics and economic compensation tracking. The platform is designed to empower the trained volunteer cadres managed by the Bihar State Disaster Management Authority (BSDMA), specifically the "Aapda Mitra" and "Ganga Prahari" units.

During a severe flood, standard commercial navigation applications become lethal liabilities, as they frequently route evacuees onto deeply submerged roadways. The Bandh Prahari platform resolves this by deploying pgRouting on the Amazon RDS PostGIS instance. The database maintains a road network graph extracted from OpenStreetMap. As the SageMaker inundation module generates the real-time flood polygon, a continuous spatial intersection query is executed against the road graph. The algorithm dynamically alters the edge cost of any road segment intersecting the floodwater to infinity (\infty), instantly pruning those routes from path calculations. This enables Aapda Mitras on the ground to generate safe, un-flooded Dijkstra shortest-path routes, guiding vulnerable populations to designated high-ground camps.

The administration of relief camps presents severe logistical challenges, compounded by the total loss of cellular connectivity. To ensure operational continuity, the platform’s relief distribution module employs a robust offline-first architecture. Field workers operate a mobile application backed by a local SQLite database. During the distribution of dry rations, medical supplies, or the registration of families for economic compensation, the volunteers scan the beneficiary's Aadhaar QR code or Ration Card barcode. Each transaction is cryptographically signed and stored locally within the SQLite queue.

When the mobile device subsequently acquires an intermittent cellular signal or connects to a satellite Wi-Fi uplink, AWS AppSync automatically executes a GraphQL delta synchronization, pushing the queued transactions to a centralized Amazon DynamoDB ledger.

Partition Key (PK)	Sort Key (SK)	Attribute Payload	Purpose
CAMP#<ID>	METADATA	{Name, LGD_Panchayat, Lat, Lon}	Camp location and jurisdictional mapping
CAMP#<ID>	ITEM#RATION	{StockOnHand: 1420, Distributed: 830}	Real-time inventory tracking
BENEFICIARY#<AadhaarHash>	RELIEF#2026_WAVE2	{FamilySize: 5, VillageLGD: "23145"}	Offline registration with conditional writes to prevent duplicate DBT claims

The DynamoDB architecture utilizes a single-table design employing conditional writes (attribute_not_exists(SK)). This mathematical constraint prevents duplicate ration collections or double-registrations across different relief camps, a pervasive issue during massive

displacement events.
Ultimately, the most critical function of the relief module is expediting the disbursement of state financial aid. The Bihar Disaster Management Department manages the distribution of ex-gratia relief (₹7,000 per affected family) directly to bank accounts via the Aapda Sampoorti portal. Delays in post-disaster manual verification surveys often stall these payments for months, disproportionately harming asset-poor households trapped in the vulnerability paradox.
The Bandh Prahari platform circumvents this delay by aggregating the verified, cryptographically stamped presence records collected by the Aapda Mitras at the relief camps. These records unequivocally prove that a family was sheltered in a designated evacuation zone. The system cross-references these registrations against the official LGD codes of the declared flood zones and automatically generates a deduplicated export file formatted to the exact schema required by the NIC Bihar Aapda Sampoorti portal. This deep integration provides District Magistrates with a pristine dataset, enabling them to authorize massive batch payouts through the Public Financial Management System (PFMS) in a fraction of the traditional timeframe, ensuring immediate economic liquidity for the state's most vulnerable populations.

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